BioNTech: The Decade Before the Pandemic

Teaching Case (HBS-style) — written for this course

For TUM Entrepreneurship Course: Prototyping Entrepreneurial Ideas for AI Module 1: Teambuilding & Why Now

How this case ties into Module 01

Read this as a **team** case, not a vaccine story. Module 01 argues that **execution compounds**: the durable advantage is rarely one brilliant idea — it is the right people executing together over years. Uğur Şahin and Özlem Türeci are the ultimate co-founder-fit story: complementary skills (a clinician-strategist and a lab-rigorous scientist), one shared conviction, and a married couple who had already built and run companies together for decades before BioNTech. Notice what carried them — **conviction before evidence**. For years mRNA was unfashionable and unproven, yet the team kept building. That is the muscle your homework trains. Your ten real conversations and your team-formation work are your version of finding the people who share your problem-conviction deeply enough to spend a decade on it. Watch how the team was built — that is the case.

Introduction

In November 2020, BioNTech and Pfizer announced that their COVID-19 vaccine was roughly 95% effective. The news sent shockwaves through the world. BioNTech's U.S.-listed shares, priced at just US$15 in its 2019 Nasdaq IPO, would climb past US$300, and the vaccine would go on to generate tens of billions of euros in revenue. Uğur Şahin, the company's co-founder and CEO, became a German national hero.

The story seemed like an overnight success.

But the truth was more complex. BioNTech had been founded in 2008, not 2020 — by Uğur Şahin, Özlem Türeci, and the veteran oncologist Christoph Huber. Şahin and Türeci had

spent the previous twelve years building technology that most of the world's investors, scientists, and pharmaceutical executives had dismissed as impractical.

Twelve years is a long time to work on an idea that nobody believes in.

This case is about how that decade unfolded. It's about a founding team with complementary skills and deep scientific conviction. It's about how timing works—not just industry timing, but team timing, technology timing, and the subtle alignment of dozens of factors. It's about what happens when founders believe in a "why now" that the market doesn't yet see.

And it's about a crucial decision in 2017: when one path seemed almost finished, should the team pivot toward an entirely different market?

Background: The Founders and the Idea

Uğur Şahin was born in 1965 in Cologne, Germany, to Turkish immigrant parents. His father was a car worker at Ford. His mother worked in a factory. His childhood was ordinary—except for one thing: his parents made him curious about science.

By age twelve, Şahin had become fascinated by tumors and cancer. He read scientific papers from his local library. He conducted experiments in the basement. By his teenage years, he had decided: he would devote his life to defeating cancer.

Özlem Türeci was born in 1967 in Marburg, Germany, also to immigrant parents (her father was a physician). She shared Şahin's obsession with oncology. She was meticulous, driven, and intensely focused on experimental rigor.

In 1990, both Şahin and Türeci entered the University of Cologne Medical School. They met in lectures. They bonded over their shared vision: cancer could be treated with precision, not blunt-force chemotherapy.

Both pursued oncology postdoctoral work. Both were trained as physicians and researchers. But by the late 1990s, they had both reached a conclusion: the traditional academic path wouldn't solve cancer. They needed to build a company.

The Message in a Molecule

Their early insight came from a simple observation about the immune system. The body's cancer-fighting machinery—T cells—can recognize tumors. But tumors are clever. They evolve to hide from the immune system. They suppress T cell activity. They create immunological "cold spots" where nothing can attack them.

What if you could wake up the immune system? What if you could teach it to recognize a specific cancer?

In 1998, a small group of scientists published a paper suggesting that messenger RNA (mRNA) could be used to instruct the immune system. Rather than injecting a cancer drug that poisons everything, you could inject instructions—an mRNA sequence—that teaches the immune system: "Here is what a cancer cell looks like. Attack it."

Theoretically, this was elegant. In practice, it had never worked in humans. mRNA was unstable. It broke down before reaching cells. The immune system treated it as a foreign invader. Dosing was unpredictable.

But Şahin and Türeci saw something that most immunologists missed. They believed that if you could solve the technical problems—stabilizing mRNA, targeting it to the right cells, getting the immune response strong enough—you would have a platform technology. Not just for cancer. For any disease where you wanted to teach the immune system to respond.

This insight would take them fifteen years to prove.

BioNTech, Founded 2008

In 2008, Şahin and Türeci — together with the veteran oncologist Christoph Huber — founded BioNTech in Mainz, Germany. Şahin became Chief Executive Officer; Türeci led the science. They hired their first scientists in a rented lab space.

What they did not have was the usual startup story. German venture capital was scarce in 2008, the financial crisis was unfolding, and biotech startups were seen as absurdly risky. But BioNTech had an unusual backer.

The Strüngmann family office — the billionaire twins Andreas and Thomas Strüngmann, who had sold their generics giant Hexal to Novartis — committed on the order of €150–180 million to the company, alongside MIG Capital and the investor Helmut Jeggle. This was patient capital, and it changed everything. Instead of scraping from grant to grant, Şahin and Türeci had anchor investors who told them, in effect, to take their time. That freedom to build quietly, away from quarterly pressure, is the reason BioNTech could spend a decade proving an idea nobody else believed in.

But they were resourceful. They recruited Sierk Pötting, a business-minded entrepreneur with pharma experience, as CFO and later CEO. They brought in Martin Bachmann, an immunologist from the University of Bern. They built a lean team of PhD scientists who were willing to work for equity and passion rather than hefty salaries.

The key insight about the team: Şahin and Türeci were not generalists. Şahin was deep in oncology and immunotherapy. Türeci was deep in molecular biology and immunology. They

needed someone who understood business operations, pharma partnerships, and capital raising. They found that in Pötting.

With that backing, the company could hire steadily and quietly. By 2010 it had dozens of employees; by 2013, more than a hundred; and by 2017, several hundred scientists and staff worked in Mainz — a substantial research organization that was still almost unknown to the outside world. The team was focused and united by a single vision: mRNA could work.

The Long March: 2008-2017

For nearly a decade, BioNTech pursued what the scientific community called “personalized cancer vaccines.” The idea was elegant:

1. Sequence a patient’s tumor (identify the mutations that make it unique)
2. Design an mRNA sequence that codes for those tumor-specific mutations
3. Inject the mRNA
4. The immune system learns to attack those mutations
5. The tumor, exposed and recognized, is defeated

On paper, this was brilliant. In practice, it was agonizingly slow.

BioNTech needed to validate the science. They ran cell culture experiments. They tested in animal models. They worked on dosing, delivery, stability. By 2012, they had published a handful of papers in decent journals. The science was credible. But credibility is not validation. The technology still hadn’t been tested in humans.

In 2014, BioNTech began its first human trials. They worked with Genmab, a larger biotech company, on melanoma. The results were modest but encouraging. The mRNA vaccine was safe. It triggered immune responses. But tumor shrinkage was limited. The approach needed refinement.

The clinical trial world is slow. In 2015, after seven years of work, BioNTech had data from dozens of patients. In 2016, they expanded trials to lung cancer and colorectal cancer. Each trial took years to enroll patients, years to generate data.

From an outside perspective, BioNTech looked stuck. The company was burning cash. It had no approved products. It had no revenue. It had promising science, but promising science fills graveyards. How many biotech startups have promising science but never make it to profitability?

The Business Model Problem

By 2016, BioNTech faced a version of biotech’s classic “valley of death.”

By 2016 the company had spent well over €150 million on R&D — most of its original backing — and still had no revenue. Clinical trials for cancer vaccines take 7-10 years. The path to profitability was at least a decade away—maybe longer.

To reach that profitability, the company needed more capital. Lots of it. But investors in cancer biotech want to see not just promising science, but validated science. They want Phase 2 trial data showing that your drug actually shrinks tumors.

BioNTech had that for melanoma. For other indications, the data was still accumulating.

Fundraising in 2016 was tough. Şahin and Türeci were not household names. Their company was based in Mainz, Germany—not Silicon Valley, not Boston. German biotech is rare. German biotech founders without pharma industry experience (both Şahin and Türeci were academics) were even rarer.

By 2016, after eight years of heavy R&D, much of the original Strüngmann commitment had been spent, and BioNTech had not yet raised a large external round from institutional investors. To keep scaling toward late-stage trials and manufacturing, it would need one — and a major financing was on the horizon.

Here's the critical point: BioNTech needed a new strategy. The cancer vaccine path was scientifically sound, but financially risky. The company needed either: (a) massive new capital from pharma partnerships, or (b) a new, faster path to validation.

The Team in 2016

By 2016, BioNTech's leadership looked like this:

Scientific founders: Uğur Şahin (CEO, oncology/immunotherapy), Özlem Türeci (Chief Medical Officer, immunology/molecular biology)

Business leader: Sierk Pötting (CFO, later co-CEO, pharma operations)

Scientific director: Martin Bachmann (immunology, vaccine design)

Chief Business Officer: Ryan Richardson (pharma business development, partnerships)

Chief Operating Officer: Katherine Janas (biotech operations, clinical trial management)

The team had evolved from just the founders and a handful of researchers. It now included operators—people who could navigate FDA processes, manage pharma partnerships, run trials.

This composition would matter enormously for what came next.

The Situation: 2017 - The Pivot Question

In early 2017, Uğur Şahin made a decision that surprised many people in the company.

He was in his early 50s now. He had spent nearly a decade on personalized cancer vaccines. The science was working—slowly. Clinical data from melanoma trials was being published in peer-reviewed journals. But the path to a marketable product was still years away.

And Şahin was thinking about a different problem.

Since the Ebola outbreak of 2014, global health organizations had been asking: what if we could develop vaccines faster? What if, instead of the traditional vaccine development cycle (5-10 years), we could develop a vaccine in weeks or months?

mRNA technology offered a path. You don't need to grow virus in eggs. You don't need to attenuate it or inactivate it. You just need to write the right genetic sequence. Theoretically, you could design an mRNA vaccine almost as fast as you could sequence the pathogen.

In 2017, there was no pandemic. There was no "why now" market. The vaccine industry was not clamoring for faster development timelines. The traditional vaccine makers—Merck, GSK, Sanofi—had settled into their decades-old processes.

But Şahin believed that pandemic preparedness was a hidden opportunity. He believed that the world was under-prepared for infectious disease. He believed that mRNA could be the solution.

In March 2017, Şahin presented this idea to the board. His proposal: BioNTech should allocate resources toward developing a platform for rapid vaccine development. The company should build partnerships with health organizations. It should pursue development of vaccines for influenza, Zika, malaria—diseases that might represent pandemic threats.

The financial case was weak. The market didn't exist yet. The timeline to validation was uncertain. But Şahin believed it was the right long-term play.

The board faced a decision:

Option A: Double down on cancer vaccines. The science was working. Clinical trials were progressing. The team was aligned. Continue the path. By 2020-2022, the first cancer vaccine might be approved. That could be worth billions.

Option B: Allocate 20-30% of resources to infectious disease vaccines. This was speculative. The market didn't exist. But if a pandemic occurred, or if regulators accelerated

vaccine development timelines, this could be valuable. It would also diversify the company's portfolio.

Option C: Pivot entirely. Abandon the cancer vaccine program (or shrink it), and commit fully to infectious disease.

Most biotech companies face this decision at least once. Most of them stick with the path they know. The institutional knowledge, the team expertise, the clinical trial infrastructure—all built for cancer. Why throw that away to pursue something uncertain?

The Team Composition Argument

But Şahin had a key insight. Looking at his team:

- Martin Bachmann had deep expertise in **vaccine immunology**—exactly what was needed for infectious disease vaccines
- Ryan Richardson had relationships across the **pharma and public health world**—he could navigate partnerships with government agencies
- Türeci had experience with **rapid translation** of basic science to clinical applications
- Pötting had **pharma operations experience**—he understood how to scale manufacturing and navigate regulatory timelines

The team wasn't just a cancer oncology shop. It was a **vaccine platform team**. The same mRNA platform that worked for cancer could work for infectious disease. The same immunological principles applied.

Şahin's argument: "We have built something bigger than a cancer company. We have built a vaccine platform company. Our team can do cancer and infectious disease. Why choose?"

The Financial Reality

In 2017, BioNTech's financial situation was precarious:

Projected Cash Runway (2017) - Cash on hand: €250 million - Monthly burn rate: €8-10 million - Runway: ~28 months

The company needed either: 1. Revenue (unlikely in the near term) 2. Major pharma partnership (possible, but with cancer vaccines only) 3. IPO or large capital raise (possible, but dependent on progress in clinical trials)

The infectious disease pivot required capital too. But it offered a different path: government grants and partnerships with health organizations. The Bill & Melinda Gates Foundation,

CEPI (Coalition for Epidemic Preparedness Innovations), and other funders were specifically interested in vaccine platform technologies.

By allocating resources to infectious disease, BioNTech could potentially unlock **new funding streams**. It could pursue grants that wouldn't apply to cancer vaccines. It could build partnerships with organizations that didn't exist yet but might emerge if there were a public health crisis.

The Decision: What Comes Next?

It is March 2017. Şahin has presented his case to the board. The conversation is intense.

The skeptical view: "We have twelve years invested in cancer vaccines. We are on the verge of real clinical validation. We have trained the world's best experts in personalized cancer immunotherapy. If we dilute our resources, we risk failure in both areas. No. We must focus."

Şahin's view: "The platform is bigger than cancer. We have a window here. The world isn't yet thinking about pandemic preparedness, but it will. We should be the company that has already solved the technical problem when the market wakes up. And if the market doesn't wake up? We still have cancer as our core business. The infectious disease work is diversification, not abandonment."

The team's role: Türeci, Bachmann, and Pötting all supported Şahin. But they understood the risk. Türeci asked: "If we dilute resources, what happens to our cancer trials? How many people does the infectious disease program need?" Pötting added: "We need a capital partner who believes in both programs. Otherwise, we're burning cash on something speculative."

The practical question: "What does this actually mean for the company? Do we hire new people? Do we partner with universities? Do we approach governments?"

Şahin's recommendation:

1. **Keep the cancer program intact**. It's the core business.
2. **Build an infectious disease program** with 15-20 people, initially focused on influenza (seasonal flu kills 290,000-650,000 people annually—a real market).
3. **Pursue government and foundation partnerships**. The US BARDA program (Biomedical Advanced Research and Development Authority), CEPI, the Gates Foundation all funded vaccine development. Approach them.

4. **Plan for a capital raise in 2018**. Use the infectious disease program as part of the investment story: "We are a vaccine platform company, not just a cancer company."

The board debated. Some directors worried about focus. Some believed Şahin's vision was overconfident. But ultimately, they agreed.

The decision: **Move forward with the dual program. Build the infectious disease capability. Maintain cancer as the core. Prepare for a 2018 capital raise.**

The Immediate Implications

If the board approves Şahin's plan, what happens next?

Hiring: The company needs 15-20 vaccine immunologists and vaccine development specialists. In Germany, these are rare. The company will need to recruit from universities, from other biotech companies, maybe from industry. This takes months. It costs money.

Partnerships: The company will approach government agencies (German, EU, US) and foundations. These relationships take time to build. Most won't commit unless they see the company is serious.

Capital: The company needs to raise at least €200 million in 2018. The investment story shifts: "We are not just an oncology company. We are a vaccine platform company with applications across cancer and infectious disease." This is more compelling to investors, but also riskier if the infectious disease program doesn't deliver.

Team dynamics: Some cancer vaccine team members may worry: "Are we deprioritized? Is the company abandoning our program?" Şahin and Türeci need to manage this carefully.

The Key Question for Students

Here's the core question the board needs to address:

When is the right time to pursue a "why now" that the market doesn't yet recognize?

Şahin believed that pandemic preparedness was inevitable. He didn't know when. He didn't know what trigger would cause the world to care. But he believed the market would eventually value rapid vaccine development.

Was he right to bet the company on this?

Exhibits

Exhibit A: BioNTech Funding to 2017

Year	Source of capital	Amount	Backers / note
2008	Founding commitment	~€150–180M committed (drawn down over the following years)	Strüngmann family office (Andreas & Thomas Strüngmann), MIG Capital, Helmut Jeggle — among the largest private commitments in European biotech
2008–2017	Public grants (non-dilutive)	—	German federal and EU research programs

Context: For its first decade BioNTech ran not on a series of venture rounds but on a single, unusually patient private commitment of roughly €150–180 million from the Strüngmann family office (with MIG Capital and Helmut Jeggle), supplemented by public grants. Moderna, founded two years later in the United States, raised capital far more publicly and in larger rounds (well over US$1 billion by 2017). BioNTech's first large external institutional round — a US$270 million Series A that valued it near US$2.3 billion — did not close until January 2018, just after the moment of this case. That patient anchor capital is why the team could build quietly for nearly ten years.

Exhibit B: BioNTech Team Composition (2017)

Role	Name	Background	Year Joined
CEO, co-founder	Uğur Şahin	Physician, oncology, immunotherapy specialist	2008
CMO, co-founder	Özlem Türeci	Physician, molecular biologist and immunologist	2008
Co-founder / Chair, Supervisory Board	Christoph Huber	Veteran oncologist, academic mentor	2008
CFO/COO	Sierk Pötting	Physicist by training; pharma finance and operations	2010
Scientific lead	Martin Bachmann	Immunologist, vaccine researcher (U. Bern)	2012
Senior Vice President (mRNA)	Katalin Karikó	mRNA biochemist (moved from UPenn)	2013
Chief Business Officer	Ryan Richardson	Pharma business development, partnerships	2015

Context: The team paired deep scientist-founders with pharma-experienced operators. By 2017, the founders had brought in people who could scale the business — crucial for the pivot.

Individuals and roles are real; specific titles and join dates are approximate.

Exhibit C: mRNA Vaccine vs. Traditional Vaccine Development Timeline

Approach	Development Time	Manufacturing Time	Total	Key Challenge
Traditional (egg-based flu)	5-6 months	3-4 months	8-10 months	Depends on fermentation
Inactivated viral (polio model)	3-4 years	6-12 months	3.5-5 years	Virus propagation
mRNA (BioNTech platform, 2017)	2-3 weeks	4-6 weeks	6-8 weeks (theoretical)	mRNA stability, delivery

Context: If mRNA vaccines worked, they could compress vaccine development timelines from years to weeks. This was the "why now"—if a pandemic emerged, you could respond in weeks instead of months.

Illustrative reconstruction for classroom use — directionally faithful, individual numbers are teaching estimates.

Exhibit D: Global Influenza Vaccine Market (2017)

Metric	Value
Annual market size	€2.2 billion
Primary buyers	Governments (CDC, ECDC, health ministries)
Annual doses produced	500-600 million
Effectiveness rate (typical)	40-60% (variable by strain match)
Unmet need	Better efficacy, faster development

Context: The influenza vaccine market was large and regulated, but dominated by incumbent manufacturers (GSK, Sanofi, Merck). An mRNA platform could potentially outperform by offering faster development and better efficacy.

Illustrative reconstruction for classroom use — directionally faithful, individual numbers are teaching estimates.

Exhibit E: Pandemic Risk Assessment (2017)

From WHO and intelligence community assessments in 2017:

Risk Factor	Assessment
Probability of pandemic in next 5 years	~25% (educated guess)
Estimated economic cost if pandemic occurs	$1+ trillion (2008 financial crisis was ~$10T, but pandemic could rival it)
Current vaccine development capacity	1-2 years minimum (too slow)
Government preparedness funding	Growing (BARDA, CEPI newly founded in 2017)

Context: By 2017, pandemic preparedness was emerging as a policy priority, but not yet urgent. The government funders existed but budgets were modest. The case for investment was not obviously compelling.

Illustrative reconstruction for classroom use — directionally faithful, individual numbers are teaching estimates.

Exhibit F: Cancer Vaccine Program Status (2017)

Melanoma Trial Results (Phase 1/2): - Patients enrolled: 97 - Median survival improvement: ~8 months vs. control - Safety profile: Generally favorable, manageable side effects - Publication timeline: Data expected in 2017-2018

Ongoing Trials: - Colorectal cancer: 127 patients, 18 months enrollment remaining - Lung cancer: 80 patients, 2+ years enrollment remaining

Timeline to FDA approval: Likely 2021-2023 (if trials successful)

Commercial potential: If approved, first mRNA cancer vaccine could be worth $1-5B annually (addressable market is ~200,000 colorectal cases/year in US alone)

Key risk: If trials fail, company has no near-term revenue

Illustrative reconstruction for classroom use — directionally faithful, individual numbers are teaching estimates.

Exhibit G: Decision Framework - Şahin's Proposal

Option A: Cancer Focus Only - Pro: Focused effort, proven team, clear path to approval - Con: Long timeline, single bet, limited funding - Outcome: Likely one approved drug by 2022, but all capital tied up - Funding need: €300M+ to get through trials

Option B: Dual Program (Cancer + Infectious Disease) - Pro: Diversification, new funding streams, larger TAM (total addressable market) - Con: Divided resources, riskier, unclear market timing - Outcome: Best case—both programs succeed. Worst case—both diluted. - Funding need: €400-500M to fund both adequately

Option C: Pivot to Infectious Disease - Pro: Faster market validation potential, government support - Con: Abandons 12 years of cancer work, losing team expertise - Outcome: Risky, likely massive mistakes in cancer field if restarted - Funding need: €250M+ for infectious disease program alone

Illustrative reconstruction for classroom use — directionally faithful, individual numbers are teaching estimates.

Exhibit H: The Timing Question - Şahin's Memo to the Board (March 2017)

[Excerpt from Şahin's written proposal]

"...The world is not yet thinking about pandemic preparedness with the urgency that the data suggests it should. But this is our opportunity. We can build the capability now, when capital is relatively available, when government interest is growing, when CEPI is launching. If we wait for a pandemic to occur, every company and institution will suddenly want this capability. We will be late.

Our mRNA platform is not just for cancer. It is for teaching the immune system. The physics of immunity doesn't change based on what disease we're trying to prevent. Our team can do this.

In 10 years, I believe vaccine companies will be the most valuable biotechs in the world, not just oncology companies. I want BioNTech to be that company."

Illustrative reconstruction for classroom use — directionally faithful, individual numbers are teaching estimates.

Discussion Questions

1. **Team composition and optionality**: How did BioNTech's team structure enable the 2017 pivot decision? What would have happened if Şahin and Türeci were purely scientists without Pötting and Richardson?
2. **"Why now" timing**: Şahin believed in pandemic preparedness in 2017, when no pandemic existed. How do founders make bets on markets that don't yet exist? What's the difference between vision and delusion?
3. **Resource allocation**: If you approve Şahin's dual-program approach, how do you actually allocate resources without destroying both programs? What would be the governance structure?
4. **Capital strategy**: How should BioNTech's fundraising pitch change if the company pursues both cancer and infectious disease? Who are the investors who believe in this thesis?
5. **Team morale and culture**: What happens to the cancer vaccine team members if infectious disease becomes a priority? How do you maintain focus and morale?
6. **Market validation**: In 2017, which infectious disease offers the fastest path to validation? Influenza? Zika? Malaria? What's the decision framework?
7. **The contrarian bet**: Most biotech founders would have said "No, let's focus on cancer." Why did Şahin push for the pivot? What does this tell us about founder psychology?
8. **Hindsight bias**: We know that a pandemic occurred in 2020 and BioNTech's vaccine was incredibly successful. Does that make Şahin's 2017 decision look obviously right? How would we evaluate the decision at the time, with no knowledge of the future?

Epilogue (read after class)

The board approved Şahin's proposal. BioNTech pursued both programs.

In January 2018, BioNTech raised a US$270 million Series A — its first large external round — from investors including Redmile Group, Janus Henderson, Invus and Fidelity, at a valuation near US$2.3 billion. The investment narrative emphasized the vaccine-platform strategy.

Through 2019, the company kept scaling. In July 2019 it raised a US$325 million Series B led by Fidelity, and in October 2019 it listed on the Nasdaq, raising about US$150 million at a valuation of roughly US$3.4 billion — still years before it had a single approved product.

In January 2020, days after the SARS-CoV-2 genome was published, BioNTech launched what it would later call Project Lightspeed. In March 2020 it partnered with Pfizer to co-develop an mRNA COVID-19 vaccine, building on an existing influenza collaboration. The cancer program continued in parallel.

In November 2020, the vaccine showed roughly 95% efficacy in its pivotal trial, and regulators cleared it within weeks. BioNTech's U.S.-listed shares — public since October 2019 — climbed past US$300 as the vaccine rolled out worldwide. The vaccine generated on the order of €19 billion in revenue in 2021 alone, and at its 2021 peak BioNTech was briefly worth more than US$90 billion — one of the most valuable biotechnology companies in Europe, and the world.

Şahin's decision to pursue "a market that didn't exist yet" became one of the most valuable bets in recent biotech history. But it only looked obvious in hindsight. In 2017, it was a reasonable but uncertain gamble.

The lesson: great founders identify inflection points before the market recognizes them. But they also maintain portfolio optionality—they don't bet everything on one uncertain future. BioNTech kept cancer vaccines going while exploring infectious disease. That balance was crucial.

Teaching case written for this course from public sources. Exhibits marked as illustrative are reconstructions for classroom use.